

EJERCICIOS DE SERIES DE POTENCIA

FORMULAS

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$y' = \sum_{n=1}^{\infty} n C_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2}$$

1. $y' = 2xy$

$$y' - 2xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 2x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} - \sum_{n=0}^{\infty} 2 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 2 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 2 C_n = 0 \quad C_1 = 0$$

$$C_{n+2} = \frac{2}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2 C_0 / 2 = C_0$$

$$n = 1 \quad C_3 = 2 C_1 / 3 = 0$$

$$n = 2 \quad C_4 = 2 C_2 / 4 = 1/2 C_0$$

Si n es impar, $C_n = 0$

$$n = 2k + 1 \quad C_n = 0 \quad C_{2k+1} = 0$$

Si n es par, $C_n = 0$

$$n = 2k \quad C_n = \frac{1}{2 * 3 * \dots * k} C_0 \quad C_{2k} = \frac{1}{2 * 3 * 4 * \dots * k} C_0 \quad C_{2k} = \frac{1}{k!} C_0$$

$$y = C_0 x^{2*0} + C_0 x^{2*1} + C_0 \frac{1}{2} x^{2*2} + C_0 \frac{1}{3} x^{2*3} + \dots \dots \dots$$

$$y = \sum_{n=0}^{\infty} \frac{1}{k!} C_0 x^{2k} = C_0 \sum_{n=0}^{\infty} \frac{(x^2)^k}{k!}$$

FORMULA

$$\sum_{n=0}^{\infty} (n+K) f(n) = \sum_{n=0}^{\infty} (n+K) f(n+K)$$

FORMULA

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots \dots \dots$$

FORMULA

$$e^v = \sum_{n=0}^{\infty} \frac{v^n}{n!}$$

$$y = C_0 * e^{x^2} \quad y = C * e^{x^2}$$

2. $xy' = (x+1)y$

$$\begin{aligned}
 xy' - xy - y &= 0 \\
 x \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=0}^{\infty} C_n x^n - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=1}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=1}^{\infty} (n+1) C_{(n+1)} x^{n+1} - \sum_{n=0}^{\infty} C_n x^{n+1} - C_0 - \sum_{n=0}^{\infty} C_{(n+1)} x^{n+1} &= 0 \\
 -C_0 + \sum_{n=0}^{\infty} [(n+1) C_{n+1} - C_n - C_{(n+1)}] x^{n+1} &= 0 \\
 (n+2) C_{n+2} - 2 C_n &= 0 \quad -C_0 = 0 \quad C_0 = 0 \\
 C_{n+1} &= \frac{C_n}{n}
 \end{aligned}$$

Remplacemos n en la ecuación obtenida

$n = 0 \quad C_1 = C_0/0 = 0$

$n = 1 \quad C_2 = C_1/1 = 0 \quad C_2 = C_1$

$n = 2 \quad C_3 = C_2/2 = 1/2 C_1$

$$C_n = \frac{1}{2 * 3 * 4 * 5 * \dots * (n-1)} C_1 \quad C_n = \frac{1}{(n-1)!} C_1$$

$$y = \sum_{n=0}^{\infty} \frac{1}{(n)!} C_1 x^{n+1} = C_1 \sum_{n=0}^{\infty} \frac{x^{n+1}}{(n)!}$$

FORMULA

$$y = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} C_1 x^n$$

$$y = \sum_{n=1}^{\infty} C_n x^n$$

$$C_1 \sum_{n=0}^{\infty} \frac{x^n}{(n)!} = C * e^{x^2}$$

3. $(2-x)y' = y$

$$\begin{aligned}
 2y' - xy - y &= 0 \\
 2 \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=0}^{\infty} C_n x^n - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=1}^{\infty} 2 C_n x^n - \sum_{n=0}^{\infty} C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 \sum_{n=0}^{\infty} 2(n+1) C_{(n+1)} x^{n+1} - \sum_{n=0}^{\infty} C_n x^{n+1} - C_0 - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
 -C_0 + \sum_{n=0}^{\infty} [2(n+1) C_{n+1} - n C_n - C_n] x^n &= 0 \\
 2(n+1) C_{n+1} - n C_n - C_n &= 0 \\
 C_{n+1} &= \frac{1}{2} C_n
 \end{aligned}$$

Remplacemos n en la ecuación obtenida

$n = 0 \quad C_1 = 1/2 C_0$

$n = 1 \quad C_2 = 1/2^2 C_0 \quad C_2 = C_1$

$$n = 2 \quad C_3 = 1/3^2 C_0$$

$$C_n = \frac{1}{2^n} C_0$$

$$y = \sum_{n=0}^{\infty} \frac{1}{2^n} C_0 x^n = C_0 \sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n$$

$$y = C_0 \frac{1}{1 - \frac{x}{2}} \quad \left| \frac{x}{2} \right| < 1$$

$$y = C_0 \frac{1}{\frac{2-x}{2}} = \frac{2C_0}{2-x}$$

$$y = \frac{C}{2-x}$$

FORMULA

$$\sum_{n=1}^{\infty} a^n = \frac{1}{1-a} \quad \text{si } |a| < 1$$

$$|x| < a \quad -a < x < a$$

$$4. \quad y' + 4xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 4x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} + \sum_{n=0}^{\infty} 4 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 4 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 4 C_n = 0 \quad C_1 = 0$$

$$C_{n+2} = \frac{4}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2 C_0$$

$$n = 1 \quad C_3 = 4/3 C_1 = 0$$

$$n = 2 \quad C_4 = 2 C_0$$

$$n = 1 \quad C_3 = 8/9 C_1 = 0$$

$$y = C_0 (2x^2 + 2x^4 + 2x^6 + \dots \dots \dots)$$

$$5. \quad (x-3)y' + 2y = 0$$

$$xy' - 3y' + 2y = 0$$

$$x \sum_{n=0}^{\infty} n C_n x^{n-1} - 3 \sum_{n=1}^{\infty} n C_n x^{n-1} + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} 3n C_n x^{n-1} + \sum_{n=0}^{\infty} 2 C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n C_n x^n - \sum_{n=0}^{\infty} 3(n+1) C_{(n+1)} x^n + \sum_{n=0}^{\infty} 2 C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [nC_n - 3(n+1)C_{n+1} + 2C_n]x^n = 0$$

$$nC_n - 3(n+1)C_{n+1} + 2C_n = 0$$

$$C_{n+1} = \frac{(n+2)}{3(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n=0 \quad C_1 = \frac{2}{3} C_0$$

$$n=1 \quad C_2 = \frac{1}{3} C_0$$

$$n=2 \quad C_3 = \frac{4}{27} C_0$$

$$C_n = \frac{n+1}{3^n} C_0$$

$$y = \sum_{n=0}^{\infty} \frac{n+1}{3^n} C_0 x^n = C_0 \sum_{n=0}^{\infty} (n+1) \left(\frac{x}{3}\right)^n$$

$$y = C_0 \frac{1}{\left(\frac{x}{3} - 1\right)^2} \quad \left|\frac{x}{3}\right| < 1$$

$$y = \frac{9C_0}{(x-3)^2}$$

FORMULA

$$\sum_{n=1}^{\infty} (n+1)x^n = \frac{1}{(x-1)^2} \quad \text{si } |1| = 1$$

$$6. \quad xy' + y = 0$$

$$x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [nC_n + C_n] x^n = 0$$

$$nC_n + C_n = 0$$

$$C_n = 0$$

$$y = \sum_{n=0}^{\infty} 0x^n = 0 \quad y = 0$$

$$7. \quad x^2 y' = y - x - 1$$

$$x^2 y' - y + x + 1 = 0$$

$$\begin{aligned}
x^2 \sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n + x + 1 &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - \sum_{n=0}^{\infty} C_n x^n &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - (C_0 + \sum_{n=1}^{\infty} C_n x^n) &= 0 \\
x + 1 - \sum_{n=0}^{\infty} n C_n x^{n+1} - C_0 - \sum_{n=1}^{\infty} C_{n+1} x^{n+1} &= 0 \\
1 - C_0 + x + \sum_{n=0}^{\infty} [-C_{n+1} + n C_n] x^{n+1} &= 0 \\
1 - C_0 + (1 - C_1)x + \sum_{n=0}^{\infty} [-C_{n+1} + n C_n] x^{n+1} &= 0 \\
1 - C_0 = 0 & \quad C_0 = 1 \\
1 - C_1 = 0 & \quad C_1 = 1 \\
C_{n+1} = n C_n &
\end{aligned}$$

Remplacemos n en la ecuación obtenida

$$n = 1 \quad C_2 = C_1$$

$$n = 2 \quad C_3 = 2 C_2$$

$$n = 3 \quad C_4 = 3 C_3$$

$$y = 1 + x + \sum_{n=2}^{\infty} (n-1)! x^n$$

$$8. \quad y'' + y = 0$$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=2}^{\infty} (n+2)(n-1) C_{n+2} x^n + \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=0}^{\infty} [(n+2)(n+1) C_{n+2} + C_n] x^n &= 0 \\
(n+2)(n+1) C_{n+2} + C_n &= 0 \\
C_{n+2} &= -\frac{1}{(n+2)(n+1)} C_n
\end{aligned}$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = -1/2 C_0$$

$$n = 1 \quad C_3 = -1/3 * 2 C_1$$

$$n = 2 \quad C_4 = 1/4 * 3 * 2 C_0$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$y = C_0 \cos x + C_1 \sin x$$

$$9. \quad y'' = 4y$$

$$y'' - 4y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n - \sum_{n=0}^{\infty} 4C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 4C_n] x^n = 0$$

$$(n+2)(n+1)C_{n+2} + C_n = 0$$

$$C_{n+2} = \frac{4}{(n+2)(n+1)} C_n$$

Reemplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 4/2 C_0$$

$$n = 1 \quad C_3 = 4/3 * 2 C_1$$

$$n = 2 \quad C_4 = 4^2/4 * 3 * 2 C_0$$

$$n = 3 \quad C_5 = 4^2/5 * 4 * 3 * 2 C_1$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{4^n x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{4^n x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{2^{2n} x^{2n}}{(2n)!} + C_1 \sum_{n=0}^{\infty} \frac{2^{2n} x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{2x^{2n}}{(2n)!} + C_1/2 \sum_{n=0}^{\infty} \frac{2x^{2n+1}}{(2n+1)!}$$

$$y = C_0 \cosh(2x) + \frac{C_1}{2} \sinh(2x)$$

FORMULA

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

FORMULA

$$\cosh(x) = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\sinh(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

10. $y'' + y = x$

$$y'' + y - x = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^n - x = 0$$

$$-x + \sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$-x + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - C_n] x^n = 0$$

$$-x + 2C_2 + C_0 + (6C_3 + C_1)x + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - C_n] x^n = 0$$

$$2C_2 + C_0 = 0$$

$$C_2 = -1/2 C_0$$

$$6C_3 + C_1 - 1 = 0$$

$$C_3 = \frac{1}{3!} C_1$$

$$(n+2)(n+1)C_{n+2} + C_n = 0$$

$$C_{n+2} = \frac{1}{(n+2)(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 2 \quad C_4 = \frac{1}{4!} C_0$$

$$n = 3 \quad C_5 = \frac{1}{5!} C_1$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} + C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$y = C_0 \cos x + C_1 \sin x + x$$

11. $y'' + 4y = 0 \quad y(0) = 0 \quad y'(0) = 3$

$$y'' + 4y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} 4C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 4C_n] x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 4C_n = 0$$

$$y(0) = 0 \quad y'(0) = 3$$

$$C_0 = 0 \quad C_1 = 3$$

$$C_{n+2} = -\frac{4}{(n+2)(n+1)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 0$$

$$n = 1 \quad C_3 = 4 * 3 / 3! C_1$$

$$n = 2 \quad C_4 = 0$$

$$n = 3 \quad C_4 = 4^2 * 3 / 5! C_1$$

$$y = \sum_{n=0}^{\infty} \frac{(-1)^n * 4^n * 3}{(2n+1)!} x^{2n+1}$$

$$y = \frac{3}{2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (2x)^{2n+1}$$

$$y = \frac{3}{2} \text{sen}(2x)$$

$$12. \quad y'' + y' - 2y = 0 \quad y(0) = 1 \quad y'(0) = 2$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2} x^n + \sum_{n=0}^{\infty} (n-1)C_{n+1} x^n - \sum_{n=0}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + (n+1)C_{n+1} - 2C_n] x^n = 0$$

$$y(0) = 1 \quad y'(0) = 2$$

$$C_0 = 1 \quad C_1 = 2$$

$$(n+2)(n+1)C_{n+2} + (n+1)C_{n+1} - 2C_n = 0$$

$$C_{n+2} = -\frac{2C_n - (n+1)C_{n+1}}{(n+2)(n+1)}$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = 2$$

$$n = 1 \quad C_3 = -4/3$$

$$n = 2 \quad C_4 = 2/3$$

$$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + \dots \dots \dots$$

$$y = 1 - 2x + 2x^2 - C_0 \frac{4}{3}x^3 + C_0 \frac{2}{3}x^4 + \dots \dots \dots$$

13. $y'' = xy$

$$y'' - yx = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$2C_2 + \sum_{n=3}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$2C_2 + \sum_{n=0}^{\infty} (n+3)(n+2)C_{n+3} x^{n+1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$2C_2 + \sum_{n=0}^{\infty} [(n+3)(n+2)C_{n+3} - C_n] x^{n+1} = 0$$

$$2C_2 = 0 \quad C_2 = 0$$

$$(n+3)(n+1)C_{n+3} + C_n = 0$$

$$C_{n+3} = -\frac{1}{(n+2)(n+3)} C_n$$

Remplacemos n en la ecuación obtenida t

$$n = 0 \quad C_3 = \frac{4}{3!} C_0$$

$$n = 1 \quad C_4 = \frac{2}{4!} C_1$$

$$n = 2 \quad C_5 = 0$$

$$n = 3 \quad C_6 = \frac{4}{6!} C_0$$

$$n = 4 \quad C_7 = \frac{2 \cdot 5}{7!} C_1$$

$$n = 5 \quad C_8 = 0$$

$$C_{3n+2} = 0 \quad C_{3n} = \frac{1 \cdot 4 \cdot 7 \dots (3n-2)}{(3n)!} C_0$$

$$C_{3n+1} = -\frac{2 \cdot 5 \dots (3n-1)}{(3n+1)!} C_1$$

$$y = C_0 \left(1 + \frac{1}{3!}x^3 + \frac{4}{6!}x^6 + \dots \right) + C_1 \left(x + \frac{1}{4!}x^4 + \dots \right)$$

14. $y'' - 2xy' + 4y = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=1}^{\infty} nC_n x^{n-1} + 4 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} 2nC_nx^n - \sum_{n=0}^{\infty} 4C_nx^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 2nC_n - 4C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 2nC_n - 4C_n = 0$$

$$C_{n+2} = -\frac{2(n-2)C_n}{(n+2)(n+1)}$$

Remplacemos n en la ecuación obtenida t

$$n=0 \quad C_2 = -2C_0$$

$$n=1 \quad C_3 = -\frac{1}{3}C_1$$

$$n=2 \quad C_4 = 0$$

$$n=3 \quad C_5 = -\frac{1}{30}C_0$$

$$y = C_0(1 - 2x^2) + C_1\left(x + \frac{1}{3}x^3 + \frac{1}{30}x^5 + \frac{1}{210}x^7 \dots\right)$$

$$15. \quad xy'' + y' + xy = 0$$

$$x \sum_{n=2}^{\infty} n(n-1)C_nx^{n-2} + \sum_{n=1}^{\infty} nC_nx^{n-1} + x \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_nx^{n-1} + \sum_{n=1}^{\infty} nC_nx^{n-1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_nx^{n-1} + C_1 + \sum_{n=2}^{\infty} nC_nx^{n-1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2)(n-1)C_{n+2}x^{n+1} + \sum_{n=0}^{\infty} (n-1)C_{n+2}x^{n+1} + \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + (n-1)C_{n+2} + C_n]x^{n+1} = 0$$

$$C_1 = 0$$

$$(n+2)(n+1)C_{n+2} + (n-1)C_{n+2} + C_n = 0$$

$$C_{n+2} = -\frac{C_n}{(n+2)^2}$$

$$n=0 \quad C_2 = -\frac{1}{2^2}C_0$$

$$n=1 \quad C_3 = 0$$

$$n=2 \quad C_4 = -\frac{1}{2^2} \cdot \frac{1}{4^2} C_0$$

$$n = 3 \quad C_5 = 0$$

$$C_{2n} = -\frac{(-1)^n C_0}{2^{2n} (n!)^2}$$

$$y = C_0 \sum_{n=2}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n} (n!)^2}$$

$$16. x^2 y'' + x^2 y' + y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + x^2 \sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) C_n x^n + \sum_{n=1}^{\infty} n C_n x^{n+1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n(n-1) C_n x^n + \sum_{n=0}^{\infty} n C_n x^{n+1} + C_0 + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$C_0 + \sum_{n=0}^{\infty} (n+1) n C_{n+1} x^{n+1} + \sum_{n=0}^{\infty} n C_n x^{n+1} + \sum_{n=0}^{\infty} C_{n+1} x^{n+1} = 0$$

$$C_0 + \sum_{n=0}^{\infty} [(n+1) n C_{n+1} + n C_n + C_{n+1}] x^{n+1} = 0$$

$$(n+1) n C_{n+1} + n C_n + C_{n+1} = 0$$

$$C_{n+1} = -\frac{n C_n}{n^2 + n + 1}$$

$$n = 0 \quad C_1 = 0$$

$$n = 1 \quad C_2 = 0$$

$$y = 0$$

$$17. 3xy'' + (2-x)y' - 2y = 0$$

$$3x \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} + 2 \sum_{n=1}^{\infty} n C_n x^{n-1} - x \sum_{n=1}^{\infty} n C_n x^{n-1} + 2 \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} 3n(n-1) C_n x^{n+1} + \sum_{n=1}^{\infty} 2n C_n x^{n+1} - \sum_{n=1}^{\infty} n C_n x^n + \sum_{n=1}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} 3n(n-1) C_n x^n + \sum_{n=0}^{\infty} 2(n+1) C_{n+1} x^n - \sum_{n=0}^{\infty} n C_n x^n + \sum_{n=0}^{\infty} 2C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [3n(n-1) C_n + 2(n+1) C_{n+1} + n C_n + 2C_n] x^n = 0$$

$$3n(n-1)C_n + 2(n+1)nC_{n+1} + nC_n + 2C_n = 0$$

$$C_{n+1} = -\frac{(n+2)C_n}{(n+1)(3n+2)}$$

$$n=0 \quad C_2 = C_0$$

$$n=1 \quad C_3 = \frac{3}{10} C_0$$

$$n=2 \quad C_4 = \frac{1}{20} C_0$$

$$y = C_0 \left(1 + x + \frac{3}{10}x^2 + \frac{1}{20}x^3 + \frac{1}{176}x^4 + \dots \right)$$

$$18. \quad y' = y$$

$$y' - y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} (n+1)C_{n+1} x^n - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+1)C_{n+1} - C_n] x^n = 0$$

$$(n+1)C_{n+1} - C_n = 0$$

$$C_{n+1} = \frac{C_n}{(n+1)}$$

$$n=0 \quad C_1 = \frac{1}{1} C_0$$

$$n=1 \quad C_2 = \frac{1}{2} C_0$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$y = C_0 e^x$$

$$19. \quad y'' - 2xy' + y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} 2nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)C_{n+2} x^n - \sum_{n=0}^{\infty} 2nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 2nC_n + C_n] x^n = 0$$

$$C_0 = 0$$

$$(n+2)(n+1)C_{n+2} - 2nC_n + C_n = 0$$

$$C_{n+2} = -\frac{(2n-1)}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = -1/2$$

$$n=1 \quad C_3 = 0$$

$$n=2 \quad C_4 = -1/8$$

$$n=4 \quad C_6 = -7/240$$

$$y = C_0 \left(1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 + \dots \right) + C_1 \left(x + \frac{1}{6}x^3 + \dots \right)$$

$$20. (x^2 + 2)y'' + 3xy' - y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 3x \sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} 3nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=1}^{\infty} 3nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} + 3nC_n - C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} + 3nC_n - C_n = 0$$

$$C_{n+2} = -\frac{(-3n-1-(n-1))}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = 1/4 C_0$$

$$n=1 \quad C_3 = -1/6 C_1$$

$$n=2 \quad C_4 = -7/96 C_0$$

$$n=4 \quad C_6 = 7/120 C_1$$

$$y = C_0 \left(1 - \frac{1}{4}x^2 - \frac{7}{96}x^4 + \dots \right) + C_1 \left(x + \frac{1}{6}x^3 + \frac{7}{120}x^5 + \dots \right)$$

$$21. (x^2 + 1)y'' - xy' + y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - x \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} + \sum_{n=0}^{\infty} nC_n x^n - \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=1}^{\infty} nC_n x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} + nC_n - C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} + nC_n - C_n = 0$$

$$C_{n+2} = -\frac{n(n-1)}{(n+2)(n+1)}C_{n+1} - \frac{(n-1)}{(n+2)(n+1)}C_n$$

$$y = 6x - 2\left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 \dots\right)$$

$$22. (x^2 + 1)y'' - 6y = 0$$

$$x^2 \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - 6 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} 2n(n-1)C_n x^{n-2} - \sum_{n=0}^{\infty} 6C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^n + \sum_{n=2}^{\infty} (n+2)(n+1)C_{n+2} x^n - \sum_{n=1}^{\infty} 6C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [n(n-1)C_n + (n+2)(n+1)C_{n+2} - 6C_n]x^n = 0$$

$$n(n-1)C_n + (n+2)(n+1)C_{n+2} - 6C_n = 0$$

$$C_{n+2} = -\frac{n(n-1)}{(n+1+6)}C_n$$

$$n=0 \quad C_1 = 0$$

$$n=1 \quad C_2 = 0$$

$$y = 0$$

$$23. y'' + xy' + 8y = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + x \sum_{n=1}^{\infty} nC_n x^{n-1} + 8 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} 8C_n x^n = 0$$

$$\sum_{n=0}^{\infty} (n+2)(n+1)C_{n+2} x^n + \sum_{n=0}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} 8C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} - 2nC_n + C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + nC_n + C_n = 0$$

$$C_{n+2} = -\frac{(-n-8)}{(n+2)(n+1)}C_n$$

$$n=0 \quad C_2 = -8/2 C_0$$

$$n=1 \quad C_3 = -9/3! C_1$$

$$n=2 \quad C_4 = 80/4! C_0$$

$$n=4 \quad C_6 = 99/5! C_1$$

$$y = C_0 \left(1 - \frac{8}{2}x^2 - \frac{8}{4!}x^4 + \dots \right) + C_1 \left(x + \frac{9}{3!}x^3 + \frac{9}{5!}x^5 \dots \right)$$

$$24. \quad y'' + 5y = 0 \quad y(0) = 0 \quad y'(0) = 1$$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + 5 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} 5C_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1)C_{n+2} + 5C_n]x^n = 0$$

$$(n+2)(n+1)C_{n+2} + 5C_n = 0$$

$$y(0) = 0 \quad y'(0) = 1$$

$$C_0 = 0 \quad C_1 = 1$$

$$C_{n+2} = -\frac{5}{(n+2)(n+1)}C_n$$

Reemplacemos n en la ecuación obtenida

$$n=0 \quad C_2 = 0$$

$$n=1 \quad C_3 = 5/3! C_1$$

$$n=2 \quad C_4 = 0$$

$$n=3 \quad C_5 = 5/5! C_1$$

$$y = C_1 \left(x + \frac{5}{3!}x^3 + \frac{5}{5!}x^5 + \frac{5}{7!}x^7 \dots \right)$$

25. $y' + 3xy = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$0 C_0 x^0 + C_1 x^{n-1} + \sum_{n=2}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} (n+2) C_{n+2} x^{n+1} + \sum_{n=0}^{\infty} 3 C_n x^{n+1} = 0$$

$$C_1 + \sum_{n=0}^{\infty} [(n+2) C_{n+2} - 3 C_n] x^{n+1} = 0$$

$$(n+2) C_{n+2} - 3 = 0 \qquad C_1 = 0$$

$$C_{n+2} = \frac{3}{(n+2)} C_n$$

Remplacemos n en la ecuación obtenida

$$n = 0 \quad C_2 = -\frac{3}{2} C_0$$

$$n = 1 \quad C_3 = 0$$

$$n = 2 \quad C_4 = -\frac{3^2}{8} C_0$$

$$n = 1 \quad C_3 = 0$$

$$y = C_0 \left(-\frac{3}{2} x^2 + -\frac{3^2}{8} x^4 + \dots \dots \dots \right)$$

26. $y' + xy = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$k = n - 1 \quad k = n + 1$$

$$n = k + 1 \quad n = k - 1$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + x \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 + \sum_{k=1}^{\infty} (k+1) C_{k+1} x^k + x \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} + C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} + C_{k-1} = 0$$

$$C_{k+1} = -\frac{C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -C_0/2$$

$$k = 2 \quad C_3 = 0$$

$$k = 3 \quad C_4 = -C_0/8$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = -C_0/48$$

$$y = C_0 \left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \frac{x^8}{384} + \dots \right)$$

$$y = C_0 \left(1 - \frac{x^2}{2^1 * 1!} + \frac{x^4}{2^2 * 2!} - \frac{x^6}{2^3 * 3!} + \frac{x^8}{2^4 * 4!} + \dots \right)$$

27. $y' - x^2 y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - x^2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^{n+2} = 0$$

$$k = n - 1 \quad k = n + 2$$

$$n = k + 1 \quad n = k - 2$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 2$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k - \sum_{k=2}^{\infty} C_{k-2} x^k = 0$$

$$C_1 + 2C_2 x + \sum_{k=2}^{\infty} (k+1) C_{k+1} x^k - \sum_{k=2}^{\infty} C_{k-2} x^k = 0$$

$$C_1 = 0$$

$$2C_2 = 0$$

$$C_2 = 0$$

$$\sum_{k=2}^{\infty} [(k+1) C_{k+1} - C_{k-2}] x^k = 0$$

$$(k+1) C_{k+1} - C_{k-2} = 0$$

$$C_{k+1} = -\frac{C_{k-2}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 2 \quad C_3 = C_0/3$$

$$k = 3 \quad C_4 = 0$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = C_0/18$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = 0$$

$$k = 8 \quad C_9 = C_0/162$$

$$y = C_0 \left(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162} + \cdots \right)$$

$$y = C_0 \left(1 + \frac{x^3}{3^1 * 1!} + \frac{x^6}{3^2 * 2!} + \frac{x^9}{3^3 * 3!} + \cdots \right)$$

28. $y' + 3x^3y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3x^3 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3 \sum_{n=0}^{\infty} C_n x^{n+3} = 0$$

$$k = n - 1 \quad k = n + 3$$

$$n = k + 1 \quad n = k - 3$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 3$$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_{k-3} x^k = 0$$

$$C_1 + 2C_2x + 3C_3x^2 + \sum_{k=3}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_{k-3} x^k = 0$$

$$C_1 = 0$$

$$2C_2x = 0$$

$$C_2 = 0$$

$$3C_3x = 0$$

$$C_3 = 0$$

$$\sum_{k=3}^{\infty} [(k+1) C_{k+1} + 3C_{k-3}] x^k = 0$$

$$(k+1) C_{k+1} + 3C_{k-3} = 0$$

$$C_{k+1} = -\frac{3C_{k-3}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 3 \quad C_4 = -\frac{3C_0}{4}$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = 0$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = \frac{9C_0}{32}$$

$$k = 8 \quad C_9 = 0$$

$$k = 9 \quad C_{10} = 0$$

$$k = 10 \quad C_{11} = 0$$

$$y = C_0 \left(1 - \frac{3}{4}x^4 + \frac{9}{32}x^8 + \dots \right)$$

$$y = C_0 \left(1 - \frac{3^1 x^4}{2^{2*1} * 1!} + \frac{3^2 x^8}{2^{2*2} * 2!} + \dots \right)$$

29. $y' + 3y = 0$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 3 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\begin{array}{cc} k = n - 1 & k = 0 \\ n = k + 1 & n = 0 \end{array}$$

Si $n = 1 \rightarrow k = 0$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 3 \sum_{k=3}^{\infty} C_k x^k = 0$$

$$\sum_{k=3}^{\infty} [(k+1) C_{k+1} + 3 C_k] x^k = 0$$

$$(k+1) C_{k+1} + 3 C_k = 0$$

$$C_{k+1} = \frac{-3 C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$k = 0 \quad C_1 = -3 C_0$

$k = 1 \quad C_2 = \frac{9}{2} C_0$

$k = 2 \quad C_3 = -\frac{9}{2} C_0$

$k = 3 \quad C_4 = \frac{27}{8} C_0$

$$y = C_0 \left(1 - 3x + \frac{9}{2} x^2 - \frac{9}{2} x^3 + \frac{27}{8} x^4 + \dots \right)$$

30. $y'' - 2xy = 0$

$$\sum_{n=2}^{\infty} (n-1)n C_n x^{n-2} - 2x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=2}^{\infty} (n-1)n C_n x^{n-2} - 2 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$\begin{array}{cc} k = n - 2 & k = n + 1 \\ n = k + 2 & n = k - 1 \end{array}$$

Si $n = 2 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$

$$\sum_{k=0}^{\infty} (k+1)(k+2) C_{k+2} x^k - 2 \sum_{k=3}^{\infty} C_{k-1} x^k = 0$$

$$2 C_2 + \sum_{k=1}^{\infty} (k+1)(k+2) C_{k+2} x^k - 2 \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$2 C_2 = 0$$

$$C_2 = 0$$

$$\sum_{k=1}^{\infty} [(k+1)(k+2) C_{k+2} - 2 C_{k-1}] x^k = 0$$

$$(k+1)(k+2) C_{k+2} - 2 C_{k-1} = 0$$

$$C_{k+2} = \frac{2C_{k-1}}{(k+1)(k+2)}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_3 = 2C_0/2 * 3$$

$$k = 2 \quad C_4 = 2C_1/3 * 4$$

$$k = 3 \quad C_5 = 0$$

$$k = 4 \quad C_6 = 2 * 2C_0/2 * 3 * 5 * 6$$

$$k = 5 \quad C_7 = 2 * 2C_1/3 * 4 * 6 * 7$$

$$k = 6 \quad C_8 = 0$$

$$k = 7 \quad C_9 = 2 * 2 * 2C_0/2 * 3 * 5 * 6 * 8 * 9$$

$$k = 8 \quad C_{10} = 2 * 2 * 2C_1/3 * 4 * 6 * 7 * 9 * 10$$

$$y = C_0 \left(1 + \frac{2}{2 * 3} x^3 + \frac{2 * 2}{2 * 3 * 5 * 6} x^6 + \frac{2 * 2 * 2}{2 * 3 * 5 * 6 * 8 * 9} x^9 + \dots \right) \\ + C_1 \left(x + \frac{2}{3 * 4} x^4 + \frac{2 * 2}{3 * 4 * 6 * 7} x^7 + \frac{2 * 2 * 2}{3 * 4 * 6 * 7 * 9 * 10} x^{10} + \dots \right)$$

$$31. \quad y' + 7xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 7x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} + 7 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$k = n - 1 \quad k = n + 1$$

$$n = k + 1 \quad n = k - 1$$

Si $n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$

$$\sum_{k=0}^{\infty} (k+1) C_{k+1} x^k + 7 \sum_{k=3}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} + 7 C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} + 7 C_{k-1} = 0$$

$$C_{k+1} = \frac{-7 C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -7C_0/2$$

$$k = 2 \quad C_3 = 0$$

$$k=3 \quad C_4 = 49C_0/8$$

$$k=4 \quad C_5 = 0$$

$$k=5 \quad C_6 = -343C_0/48$$

$$k=6 \quad C_7 = 0$$

$$k=7 \quad C_8 = 2401C_0/384$$

$$k=8 \quad C_9 = 0$$

$$y = C_0(1 - \frac{7}{2}x^2 + \frac{49}{8}x^4 - \frac{343}{48}x^6 + \frac{2401}{384}x^8 + \dots)$$

$$32. \quad y' + 2y = 0$$

$$\begin{aligned} \sum_{n=1}^{\infty} nC_n x^{n-1} + 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\ \sum_{n=0}^{\infty} (n+1)C_{n+1} x^n + 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\ \sum_{n=0}^{\infty} [(n+1)C_{n+1} + 2C_n] x^n &= 0 \\ (n+1)C_{n+1} + 2C_n &= 0 \\ C_{n+1} &= \frac{-2C_n}{n+1} \end{aligned}$$

Remplacemos k en la ecuación obtenida

$$n=0 \quad C_1 = -2C_0/1$$

$$n=1 \quad C_2 = -2C_1/2 = 2^2 C_1/2!$$

$$n=2 \quad C_3 = -2C_2/3 = 2^3 C_1/3!$$

$$C_n = (-1)^n \frac{2^n C_0}{n!}$$

$$y = \sum_{n=0}^{\infty} (-1)^n \frac{2^n C_0}{n!} x^n$$

$$y = C_0 \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!}$$

$$y = C_0 e^{-2x}$$

$$33. \quad y'' - 2xy' - 2y = 0$$

$$\sum_{n=0}^{\infty} n(n-1)C_n x^{n-2} - 2x \sum_{n=0}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\begin{aligned}
& \sum_{n=0}^{\infty} n(n-1)C_n x^{n-2} - 2 \sum_{n=0}^{\infty} nC_n x^n - 2 \sum_{n=0}^{\infty} C_n x^n = 0 \\
& \sum_{n=0}^{\infty} (n+2)(n-1)C_n + 2x^n - 2 \sum_{n=0}^{\infty} nC_n x^n - 2 \sum_{n=0}^{\infty} C_n x^n = 0 \\
& (n+2)(n-1)C_{n+2} - 2nC_n - 2C_n = 0 \\
& C_{n+2} = \frac{2(n+1)C_n}{(n+2)(n+1)} = \frac{2C_n}{n+2}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$n = 0 \quad C_2 = C_n$$

$$n = 1 \quad C_3 = 2C_1/3$$

$$n = 2 \quad C_4 = C_0/2$$

$$n = 3 \quad C_5 = 0$$

$$n = 4 \quad C_6 = C_0/6$$

$$y = C_0 \left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots \right) + C_1 \left(x + \frac{2x^3}{3} + \frac{2^2 x^5}{3 \cdot 5} + \dots \right)$$

$$34. \quad xy'' + (x+1)y = 1$$

$$\begin{aligned}
& x \sum_{n=2}^{\infty} (n-1)nC_n x^{n-2} + (x+1) \sum_{n=0}^{\infty} C_n x^n = 1 \\
& x \sum_{n=2}^{\infty} (n-1)nC_n x^{n-2} + x \sum_{n=0}^{\infty} C_n x^n + 1 \sum_{n=0}^{\infty} C_n x^n = 1 \\
& \sum_{n=2}^{\infty} (n-1)nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} + \sum_{n=0}^{\infty} C_n x^n = 1 \\
& \quad \begin{array}{ccc} k = n-1 & k = n+1 & k = n \\ n = k+1 & n = k-1 & k = 0 \end{array} \\
& \text{Si } n = 2 \rightarrow k = 1 \quad n = 0 \rightarrow k = 1 \\
& \sum_{k=1}^{\infty} k(k+1)C_{k+1}x^k + \sum_{k=1}^{\infty} C_{k-1}x^k + C_0 + \sum_{k=0}^{\infty} C_k x^k = 1 \\
& \quad C_0 = 0 \\
& \sum_{n=0}^{\infty} [k(k+1)C_{k+1} + C_{k-1} + C_k] x^n = 0 \\
& \quad (k+1)C_{k+1} + C_{k-1} + C_k = 0 \\
& \quad C_{k+1} = \frac{-C_{k-1} - C_k}{(k+1)k}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = -C_1/2 - 1/2$$

$$k = 2 \quad C_3 = -C_1/12 + 1/12$$

$$k = 3 \quad C_4 = 7/144 C_1 - 5/144$$

$$y = C_0 + C_1 x + \left(-\frac{C_1}{2} - \frac{1}{2}\right) x^2 + \left(-\frac{C_1}{12} + \frac{1}{12}\right) x^3 + \left(\frac{7}{144} C_1 - \frac{5}{144}\right) x^4 + \dots$$

$$y = C_0 + C_1 \left(x - \frac{1}{2} x^2 - \frac{1}{12} x^3 + \frac{7}{144} x^4 + \dots \right) - \frac{1}{2} x^2 + \frac{1}{12} x^3 - \frac{5}{144} x^4 + \dots$$

$$35. \quad y' - 4xy = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 4x \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 4 \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$k = n - 1 \quad k = n + 1$$

$$n = k + 1 \quad n = k - 1$$

$$\text{Si } n = 1 \rightarrow k = 0 \quad n = 0 \rightarrow k = 1$$

$$C_1 + \sum_{k=0}^{\infty} (k+1) C_{k+1} x^k - 4 \sum_{k=1}^{\infty} C_{k-1} x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=1}^{\infty} [(k+1) C_{k+1} - 4 C_{k-1}] x^k = 0$$

$$(k+1) C_{k+1} - 4 C_{k-1} = 0$$

$$C_{k+1} = \frac{4 C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 1 \quad C_2 = 2 C_0$$

$$k = 2 \quad C_3 = 0$$

$$k = 3 \quad C_4 = 2 C_0$$

$$k = 4 \quad C_5 = 0$$

$$k = 5 \quad C_6 = 4 C_0 / 3$$

$$k = 6 \quad C_7 = 0$$

$$k = 7 \quad C_8 = 2 C_0 / 3$$

$$k = 8 \quad C_9 = 0$$

$$y = C_0 \left(1 + 2x^2 + 2x^4 + \frac{4}{3} x^6 + \frac{2}{3} x^8 + \dots \right)$$

36. $y'' - xy' + 2y = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} nC_n x^n + 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)C_k + 2x^k - \sum_{k=1}^{\infty} kC_k x^k + 2 \sum_{k=0}^{\infty} C_k x^k$$

$$2C_2 + 2C_0 + \sum_{k=0}^{\infty} [(k+2)(k+1)C_k + 2 - (k-2)C_k]x^k = 0$$

$$2C_2 + 2C_0 = 0$$

$$(k+2)(k+1)C_k + 2 - (k-2)C_k = 0$$

$$C_2 = -C_0$$

CASO 1, $C_0 = 1$ y $C_1 = 0$

$$C_2 = -1$$

$$C_3 = C_5 = C_7 = 0$$

$$C_4 = 0$$

$$C_6 = C_8 = C_{10} = 0$$

CASO 2, $C_0 = 0$ y $C_1 = 1$

$$C_2 = C_4 = C_6 = 0$$

$$C_3 = -1/6$$

$$C_5 = -1/120$$

$$y_1 = 1 - x^2$$

$$y_2 = x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \dots$$

37. $y'' + x^2y' + xy = 0$

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+2)(k+1)C_k + 2x^k + \sum_{k=2}^{\infty} (k+1)C_k - 1x^k + \sum_{k=1}^{\infty} C_{k-1}x^n$$

$$2C_2 + (6C_3 + C_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)C_{k+2} + kC_{k-1}]x^k = 0$$

$$C_2 = 6C_3 + C_0 = 0$$

$$(k+2)(k+1)C_k + kC_{k-1} = 0$$

$$C_3 = -1/6 C_0$$

$$C_{k+2} = \frac{k}{(k+2)(k+1)} C_{k-1}$$

CASO 1, $C_0 = 1$ y $C_1 = 0$

$$C_2 = -1/6 C_0$$

$$C_4 = C_5 = 0$$

$$C_6 = -1/45$$

CASO 2, $C_0 = 0$ y $C_1 = 1$

$$C_3 = 0$$

$$C_4 = -1/6$$

$$C_5 = C_6 = 0$$

$$C_7 = 5/252$$

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{45}x^6 - \dots$$

$$y_2 = x - \frac{1}{6}x^4 + \frac{5}{232}x^7 - \dots$$

$$38. y' + y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n - 1 \quad k = n$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + C_k]x^k = 0$$

$$(k+1)C_{k+1} + C_k = 0$$

$$C_{k+1} = -\frac{C_k}{k+1} C_{k-1}$$

Remplacemos k en la ecuación obtenida

$$k = 0 \quad C_1 = -C_0$$

$$k = 1 \quad C_2 = C_0/2$$

$$k = 2 \quad C_3 = -C_0/6$$

$$k = 3 \quad C_4 = C_0/24$$

$$k = 4 \quad C_5 = -C_0/120$$

$$y = C_0(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!})$$

$$39. y' - 2y = 0$$

$$\sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$k = n - 1 \quad k = n$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k - 2 \sum_{k=0}^{\infty} C_k x^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} - 2C_k]x^k = 0$$

$$(k+1)C_{k+1} - 2C_k = 0$$

$$C_{k+1} = \frac{2C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k = 0 \quad C_1 = 2C_0$$

$$\begin{aligned}
k=1 & \quad C_2 = 2C_0 \\
k=2 & \quad C_3 = 4C_0/3 \\
k=3 & \quad C_4 = 8C_0/12 \\
k=4 & \quad C_5 = 16C_0/60
\end{aligned}$$

$$y = C_0(1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{8}{12}x^4 + \frac{16}{60}x^5)$$

40. $y' - x^2y = 0$

$$\begin{aligned}
\sum_{n=1}^{\infty} nC_n x^{n-1} - x^2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{k=0}^{\infty} nC_n x^{n-1} - \sum_{k=0}^{\infty} C_k x^{k+2} &= 0 \\
C_1 + C_2 x + \sum_{n=3}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^{n+2} &= 0 \\
C_1 &= 0 \\
C_2 &= 0 \\
\sum_{k=0}^{\infty} (k+3)C_{k+3} x^{k+2} - \sum_{n=0}^{\infty} C_n x^{n+2} &= 0 \\
\sum_{k=0}^{\infty} [(k+3)C_{k+3} - C_k] x^{k+2} &= 0 \\
(k+3)C_{k+3} - C_k &= 0 \\
C_{k+3} &= \frac{C_k}{k+3}
\end{aligned}$$

Remplacemos k en la ecuación obtenida

$$\begin{aligned}
k=0 & \quad C_3 = C_0/3 \\
k=1 & \quad C_4 = 0 \\
k=2 & \quad C_5 = 0 \\
k=3 & \quad C_6 = C_0/18 \\
k=4 & \quad C_7 = 0 \\
k=5 & \quad C_8 = 0 \\
k=6 & \quad C_9 = C_0/162
\end{aligned}$$

$$y = C_0(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162})$$

41. $y'' + y' - 2y = 0 \quad y(0) = 1, y'(0) = 2$

$$\begin{aligned}
\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - 2 \sum_{n=0}^{\infty} C_n x^n &= 0 \\
\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sum_{n=1}^{\infty} nC_n x^{n-1} - \sum_{n=0}^{\infty} 2C_n x^n &= 0
\end{aligned}$$

$$\sum_{n=2}^{\infty} (n+2)(n-1)C_{n+2}x^n + \sum_{n=0}^{\infty} (n+1)C_{n+1}x^n - \sum_{n=0}^{\infty} 2C_nx^n = 0$$

$$\sum_{k=0}^{\infty} [(n+2)(n-1)C_{n+2} + (n+1)C_{n+1} - 2C_n]x^n = 0$$

$$(n+2)(n-1)C_{n+2} + (n+1)C_{n+1} - 2C_n = 0$$

$$(n+2)(n-1)C_{n+2} = -(n+1)C_{n+1} + 2C_n$$

$$C_{n+2} = \frac{2C_n - (n+1)C_{n+1}}{(n+2)(n+1)}$$

Remplacemos k en la ecuación obtenida

$$n=0 \quad C_2 = 0$$

$$n=1 \quad C_3 = -4/3$$

$$n=2 \quad C_4 = 2/3$$

$$n=3 \quad C_5 = -4/15$$

$$n=4 \quad C_6 = 4/45$$

$$n=5 \quad C_7 = -8/315$$

$$n=6 \quad C_8 = 2/315$$

$$y = C_0(1 + \frac{x^3}{3} + \frac{x^6}{18} + \frac{x^9}{162})$$

$$42. \quad y' + 3y = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 3 \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + 3 \sum_{k=0}^{\infty} C_kx^k = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + 3C_k]x^k = 0$$

$$(k+1)C_{k+1} + 3C_k = 0$$

$$C_{k+1} = \frac{-3C_k}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k=0 \quad C_1 = -3C_0$$

$$k=1 \quad C_2 = 9/2 C_0$$

$$k=2 \quad C_3 = -9/2 C_0$$

$$k=3 \quad C_4 = 27/8 C_0$$

$$y = C_0(1 - 3x + \frac{9}{2}x^2 - \frac{9}{2}x^3 + \frac{27}{8}x^4 + \dots)$$

$$43. \quad y' + 7xy = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 7x \sum_{n=0}^{\infty} C_nx^n = 0$$

$$\sum_{n=1}^{\infty} nC_nx^{n-1} + 7 \sum_{n=0}^{\infty} C_nx^{n+1} = 0$$

$$\sum_{k=0}^{\infty} (k+1)C_{k+1}x^k + 7 \sum_{k=1}^{\infty} C_{k-1}x^k = 0$$

$$C_1 + \sum_{k=1}^{\infty} (k+1)C_{k+1}x^k + 7 \sum_{k=1}^{\infty} C_{k-1}x^k = 0$$

$$C_1 = 0$$

$$\sum_{k=0}^{\infty} [(k+1)C_{k+1} + 7C_{k-1}]x^k = 0$$

$$(k+1)C_{k+1} + 7C_{k-1} = 0$$

$$C_{k+1} = \frac{-7C_{k-1}}{k+1}$$

Remplacemos k en la ecuación obtenida

$$k=1 \quad C_2 = -\frac{7C_0}{2}$$

$$k=2 \quad C_3 = 0$$

$$k=3 \quad C_4 = \frac{49C_0}{8}$$

$$k=4 \quad C_5 = 0$$

$$k=5 \quad C_6 = -\frac{343C_0}{48}$$

$$k=6 \quad C_7 = 0$$

$$k=7 \quad C_8 = \frac{2401C_0}{384}$$

$$k=8 \quad C_9 = 0$$

$$y = C_0 \left(1 - \frac{7}{2}x^2 + \frac{49}{8}x^4 - \frac{343}{48}x^6 + \frac{2401}{384}x^8 \dots \right)$$